

Hospital Network System

Using Cisco Packet Tracer

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1. Introduction

This project presents the design and simulation of a hospital network infrastructure integrated with IoT devices using Cisco Packet Tracer. The network is divided into multiple departments using VLAN technology to improve security, performance, and network management. An IoT-based fire alert system consisting of a Smoke Detector and Siren is also integrated to demonstrate smart healthcare infrastructure.

2. Objectives

- Design a secure and efficient hospital network.
- Implement VLANs for departmental segregation.
- Configure inter-VLAN communication using Router-on-a-Stick.
- Provide automatic IP addressing using DHCP.
- Integrate IoT devices into the network.
- Simulate a fire detection and alert system.
- Test and verify network connectivity.

3. Network Design

The hospital network is divided into three departments:

Department	VLAN ID	Network
Reception	VLAN 10	192.168.10.0/24
Admin	VLAN 20	192.168.20.0/24
Doctors	VLAN 30	192.168.30.0/24

Devices Used

- Cisco Router
- Core Switch
- Access Switch
- PCs
- Home Gateway

- Smoke Detector
- Siren
- Registration Server
- Ethernet Cables

4. Network Configuration

VLAN Configuration

Three VLANs were created on the switch:

- VLAN 10 – Reception
- VLAN 20 – Admin
- VLAN 30 – Doctors

Each department's devices were assigned to their respective VLANs.

Trunk Configuration

A trunk link was configured between the switch and router to allow communication among VLANs.

Inter-VLAN Routing

Router-on-a-Stick configuration was implemented using router subinterfaces:

```
192.168.10.1
192.168.20.1
192.168.30.1
```

These gateways enabled communication between departments.

DHCP Configuration

DHCP was configured to automatically assign IP addresses to network devices.

5. IoT Integration

IoT Devices

- Smoke Detector
- Siren
- Home Gateway

The IoT devices were connected to the Home Gateway through wireless connectivity.

IP Address Assignment

The Home Gateway provided IP addresses to IoT devices through DHCP:

```
Smoke Detector : 192.168.25.102
Siren : 192.168.25.100
```

Registration Server

A Registration Server was added and connected to the Core Switch.

```
IP Address : 192.168.10.200
```

```
Subnet Mask : 255.255.255.0
```

```
Default Gateway: 192.168.10.1
```

Network connectivity between the server and other devices was successfully verified.

6. Testing and Verification

VLAN Testing

The following command was used:

```
show vlan brief
```

VLANs were successfully created and assigned.

Connectivity Testing

Ping tests were performed between different VLANs:

```
ping 192.168.20.x
```

```
ping 192.168.30.x
```

Successful replies received.

Server Connectivity Testing

```
ping 192.168.10.200
```

Reply received successfully.

Smoke Detector Testing

The Smoke Detector state was changed from State = 0 to State = 1. The device indicator changed to red, confirming event detection.

7. Challenges Faced

- IoT automation between the Smoke Detector and Siren could not be fully implemented.
- Although device connectivity was successful, Packet Tracer 9.0.0 did not display the IoTautomation interface properly.
- The Registration Server remained limited despite successful network connectivity.

8. Results

Successfully implemented:

- VLAN Segmentation
- Inter-VLAN Routing
- DHCP Services
- Network Connectivity Testing
- Home Gateway Configuration
- IoT Device Connectivity
- Registration Server Configuration

9. Conclusion

The Hospital Network and IoT-Based Fire Alert System was successfully designed and implemented in Cisco Packet Tracer. The network infrastructure including VLANs, routing, DHCP, and device connectivity functioned correctly. IoT devices were successfully integrated into the network and event detection was demonstrated using the Smoke Detector. The project illustrates how modern healthcare facilities can utilize networking and IoT technologies to improve operational efficiency and safety.

10. Future Improvements

- Implement fully functional IoT automation.
- Add CCTV monitoring systems.
- Integrate email/SMS alerts for emergencies.
- Deploy firewall and advanced security policies.
- Implement centralized network monitoring and management.